House prices in Jabodetabek: 22.7% typical error, against 28.0% for the area median

Model report: how it was built, scored once on 562 held-out listings, and where it fails

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The model beats the best baseline by 5.3 points

Result. Median error of price is 22.7% on 562 listings scored once (95% interval 20.9% to 25.1%), against 28.0% for the kecamatan median (p. 7).

What carries it. Extra trees carries the ensemble; XGBoost adds nothing measurable (p. 8). A score from the listing text is the strongest input (p. 9).

Where it fails. Outer and cheap areas: Kab. Bogor and Kota Bogor are the worst cities, and the cheapest quarter of listings is over-estimated (p. 10, 11).

Honest ranges. The 80% range held 80.6% of holdout listings (p. 12). The checks found little leakage (p. 13).

Limits. These are asking prices, not sale prices, and 23 of 183 kecamatan have fewer than 5 listings (p. 14).

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We judge the model by its median error on 562 listings it never saw

A house listed at Rp 1.00 bn is typically estimated between Rp 0.77 and 1.23 bn. The holdout was scored once, after every choice.

22.7%

median error: half of the 562 held-out houses are estimated closer than this

45%

of estimates fall within 20% of the asking price (253 of 562)

34.3%

mean error: a few big misses lift it. On training folds the worst 10% carry 40% of all error

Source: final holdout scoring, reports/tables/final_test.csv. Tail share: error_analysis_tail.csv (training folds).

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The model learns from 4,031 Home Listing Web house listings, and they skew toward places named in the listing

Only still-active listings are in the data, so older months hold fewer rows and slow-selling houses may be over-represented.

What a row is. One active house listing in Jabodetabek, created 2025-04 to 2026-08. 3,469 rows train the model; 562 were set aside before modelling and scored once.

What the price is. The asking price. Asking prices run above what houses sell for, so every error here is against an asking price.

How rows were chosen. Listings were found by place names in their web address, so houses whose address names only a street or a cluster are under-represented.

What is added. Distances to rail, MRT/LRT, busway and Monas, flood hazard, land-use around the site and the kecamatan, from OpenStreetMap, BNPB InaRISK and BIG boundaries.

Source: Home Listing Web listings (aggregates only), OpenStreetMap (ODbL), BIG, BNPB; counts from audit_same_house_leakage.csv.

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Judged on unseen areas, the error rises from 20.1% to 23.1%

We pick models by the spatial scheme, because the public map must also work for areas with few listings.

How it is tested	Rows	Median error
Random 5x5 cross-validation (neighbours may be in training)	3,896	20.1%
Spatial-block cross-validation (whole areas held out)	**3,896**	**23.1%**
Holdout, set aside before modelling, scored once	562	22.7%

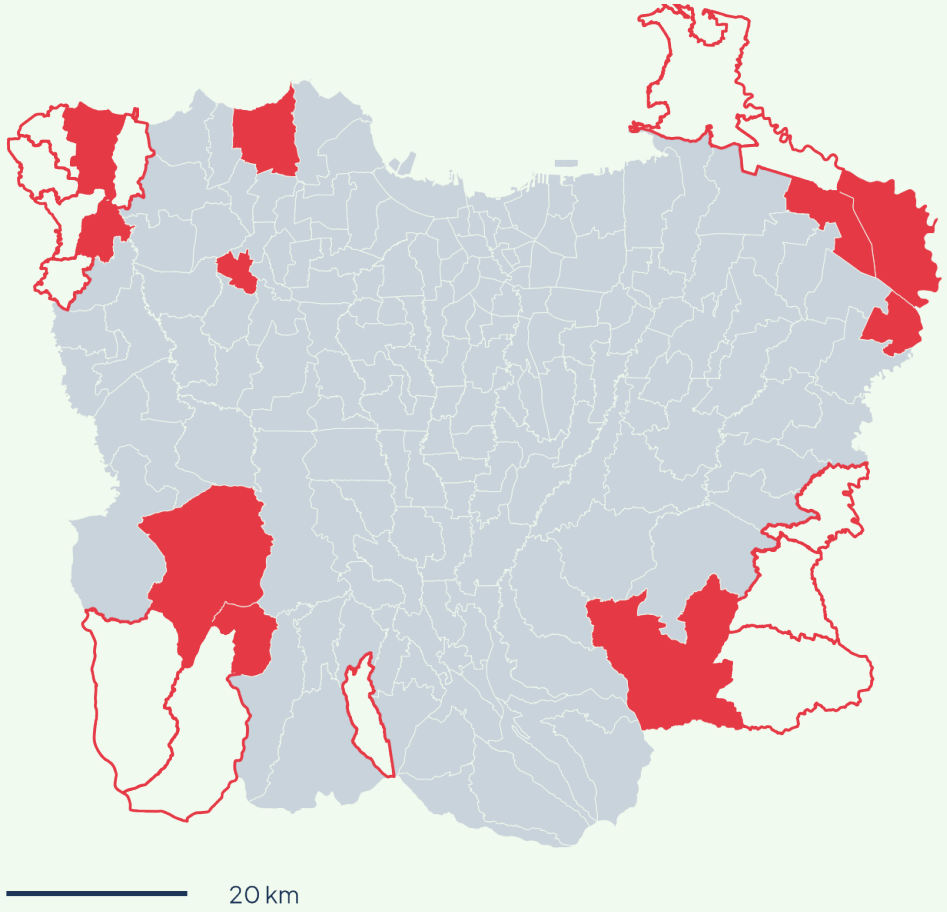
Note: The cross-validation rows are the 3,896 usable listings at the 2026-11-08 run, slightly newer than the 3,469 the model was fitted on.

Source: experiments.csv (E25 reference, 5 repeats); final_test.csv.

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23 of 183 kecamatan have fewer than 5 listings, so estimates there extrapolate

13 kecamatan have none and 10 have 1 to 4; the other 160 have 5 or more.



5 or more listings
1 to 4 listings
no listings

Where. The thin areas are remote parts of Kab. Bogor, Kab. Tangerang and Kab. Bekasi.

Why it matters. Under spatial-block testing the error stops falling after about 2,000 rows (appendix, p. 17): more rows help known areas, not unseen ones.

Source: BIG kecamatan boundaries (EPSG:32748 shown); listing counts from kecamatan_prices.csv (4,029 usable listings).

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The ensemble cuts the error by 5.3 points against the kecamatan median

The previous tuned XGBoost scored 25.1%; its interval overlaps the ensemble's.

Model	Median error	95% interval	Within 20%
City median	35.8%	31.5% to 39.8%	30%
Kecamatan median	28.0%	25.5% to 29.6%	36%
Linear regression	27.5%	23.9% to 29.3%	38%
Tuned XGBoost (previous model)	25.1%	22.2% to 28.8%	43%
Histogram gradient boosting	24.8%	21.7% to 27.9%	42%
Ensemble (final)	**22.7%**	**20.9% to 25.1%**	**45%**

Note: Intervals are bootstrap intervals on 562 listings; the ensemble's and the tuned XGBoost's overlap, so that gap is thin.

Source: final holdout scoring, reports/tables/final_test.csv.

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Dropping XGBoost moves the error by about 0.1 point, so extra trees carries the ensemble

XGBoost is kept for now: a change needs a fresh holdout, and the current one is spent.

Variant	Random CV	Spatial CV
Final ensemble (reference)	20.1%	23.1%
Without ridge	19.9%	23.4%
Without extra trees	21.0%	23.3%
Without XGBoost	**20.2%**	**23.1%**
XGBoost alone	21.0%	23.9%
Extra trees alone	19.9%	23.6%
Ridge alone	22.5%	24.4%

Note: Medians over repeats; differences under about 0.3 points are within the spread between repeats.

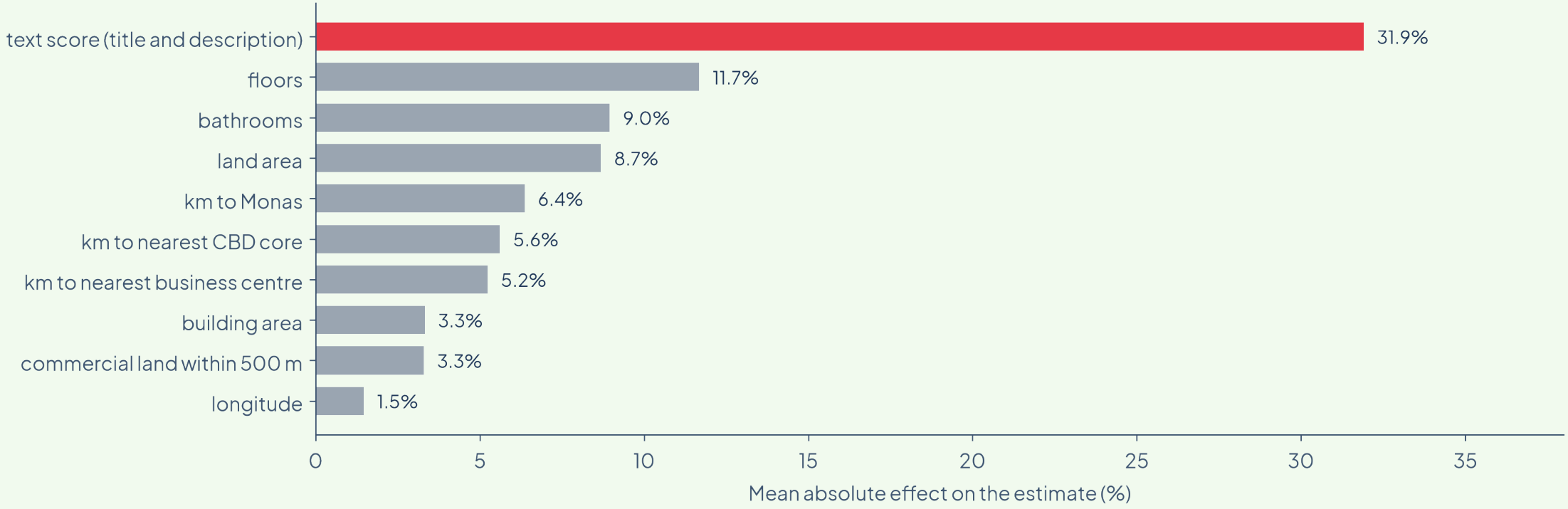
Source: experiments.csv, E25 ablation on 3,896 listings, 5x5 cross-validation.

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A score from the listing's title and description is the strongest input: it moves the estimate by 32% on average

It mostly encodes place names and wording such as "mewah", "subsidi" or "murah", learned from full descriptions. Floors and bathrooms come next.

Ten largest inputs by mean absolute effect on the estimate (%), SHAP



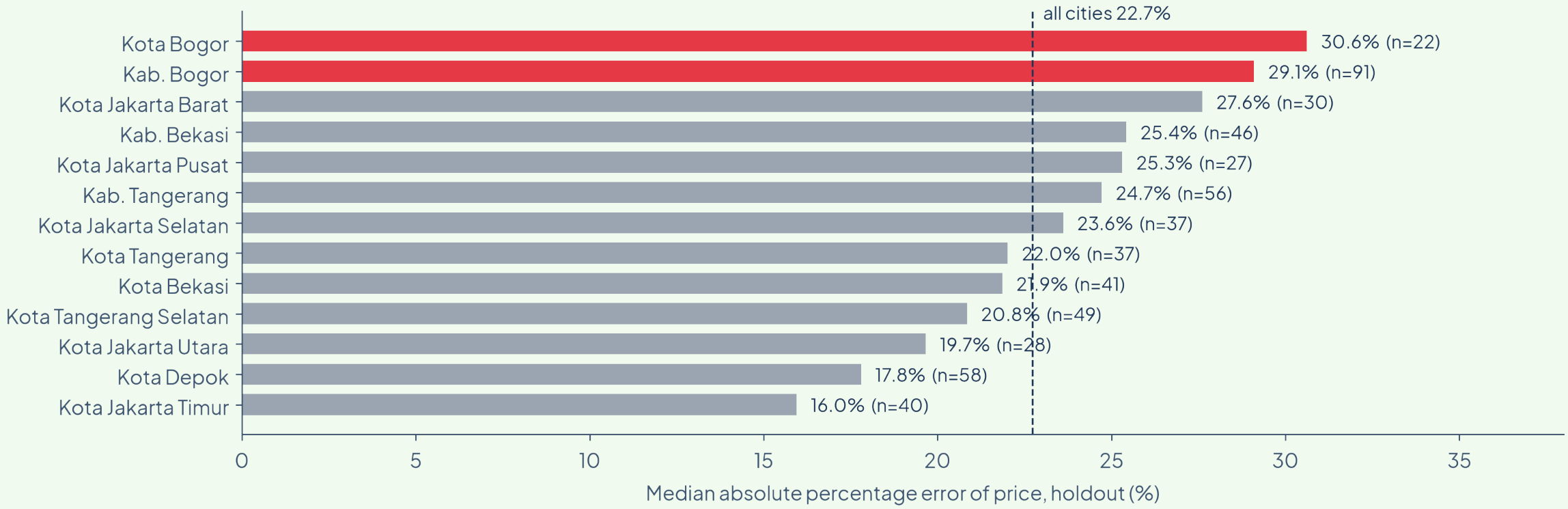
Source: model_shap_importance.csv (SHAP on the fitted model; model output, not causal).

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Errors run from 16.0% in Kota Jakarta Timur to 30.6% in Kota Bogor

Each city has only 22 to 91 holdout listings, so the order is rough; the two worst are marked.

Median absolute percentage error of price by city, holdout (%)



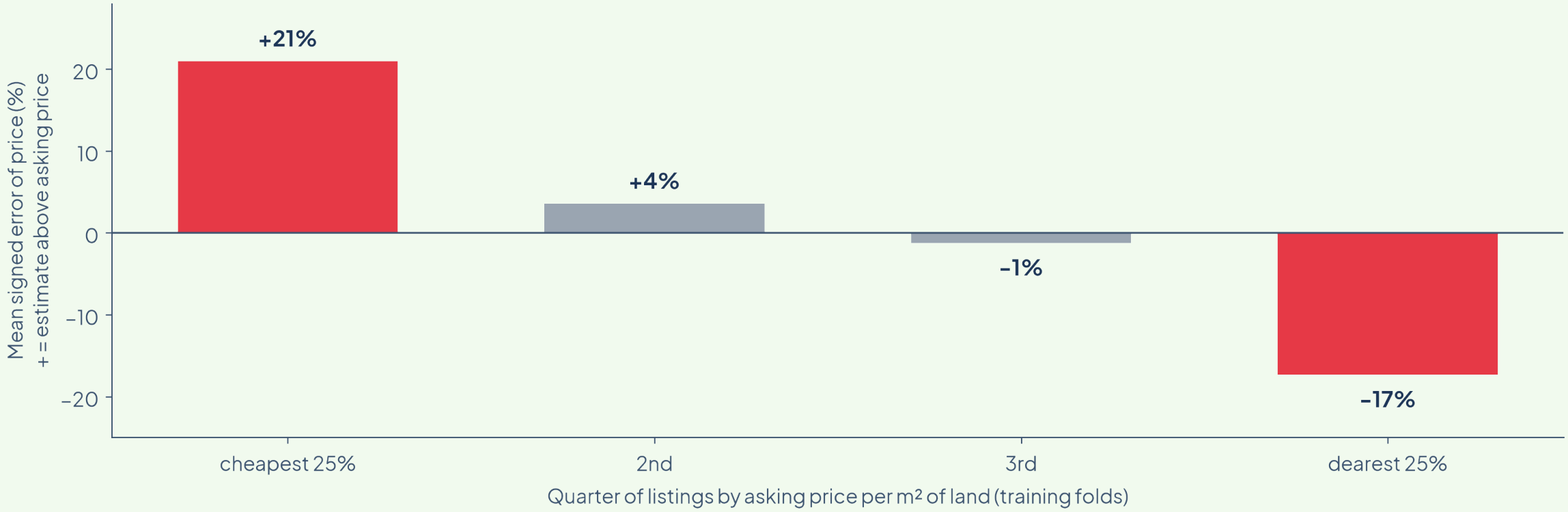
Source: final_test_segments.csv.

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The cheapest quarter is over-estimated by 21% and the dearest under-estimated by 17%

This pull toward the middle is regression to the mean; houses over 300 m² of land are also hard: 31.8% median error.

Mean signed error of price by quarter of asking price per m² of land (%), training folds



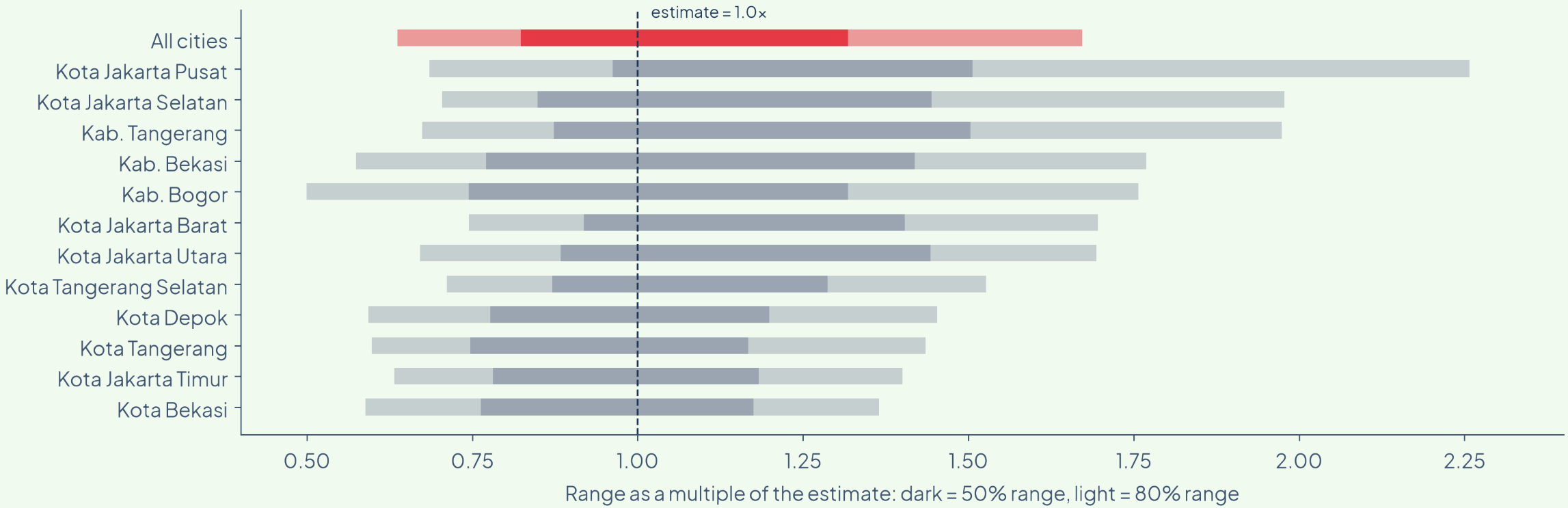
Source: error_analysis.csv (out-of-fold on the training listings).

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The 80% range of 0.64x to 1.67x held 80.6% of holdout listings

The 50% range held 52.3% (target 50%); ranges are widest in Jakarta Pusat, Jakarta Selatan and Kab. Tangerang.

Ranges around an estimate, from spatial out-of-fold errors, by city (multiple of the estimate)



Source: prediction_ranges.csv; final_test_range_coverage.csv.

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Removing the 43 same-house listings moves the holdout error from 22.8% to 22.9%

The audit recomputes the holdout score at 22.8%; the headline 22.7% is from the single final scoring.

Same house twice. 43 holdout listings have the same house in training. Without them the holdout error is 22.9% (21.1% to 25.4%); those 43 alone score 18.1%. The leak is small.

Agent noise. The same house listed by different agents differs by about 6% in price (from 289 pairs), small next to the model error.

The listing site's own estimate. On 3,291 training listings, the listing site's estimate has a median error of 23.4% against 20.1% for this model. On listings it does not flag as low accuracy the two are level (15.1% against 15.8%). Whether its model saw the asking price is unknown.

Source: audit_same_house_leakage.csv, noise_floor.csv, audit_portal_benchmark.csv.

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What the model cannot do: it knows asking prices and place, not the condition of the house

Asking, not sale prices. The estimate is not an appraisal, a mortgage valuation or advice. Asking prices run above what houses sell for.

Unseen quality. Condition, finish, street width and exact position are not captured; no photos are used.

Coarse locations. About a third of listings gave no coordinate of their own, only the district centre, and the model learned from them.

Thin areas. 23 of 183 kecamatan have fewer than 5 listings; estimates there extrapolate and the range is wider.

Time. Trained on listings from 2025-04 to 2026-08; prices drift, so the model is re-trained monthly and its live error is published.

Source: model card, limitations section; kecamatan counts from kecamatan_prices.csv.

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The model is live as a free public map and estimator, with a monthly gate before any new model replaces it

Shipped. A free web map of price per m² of land by kecamatan (shown only where 5 or more listings exist) and a price estimate with a usual range. Only aggregates and estimates are public.

How it stays honest. A weekly job scores the live model on listings first seen after training; a monthly challenger replaces it only if it is better than the noise on identical folds and in no city worse by a set margin.

Open. The weekly run of 2026-11-09 flagged a bias of +13.4% in Kota Bekasi; the cause is open, and the model is unchanged until the next monthly gate.

Waits for a fresh holdout. Any change to the point model, including dropping XGBoost, because the current holdout is spent.

Source: model card (monitoring section); project decisions log.

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Appendix: terms used in this report

Median error. Take each house's error as a share of its asking price, line them up, and read the middle one. Half are closer, half are further.

Mean error. The same errors averaged. A few large misses lift it above the median.

Holdout. 562 listings set aside before modelling and scored once at the end, so they cannot have shaped any choice.

Cross-validation. Repeatedly hiding part of the training data, predicting it, and scoring the guesses.

Spatial-block CV. Cross-validation that hides whole neighbourhoods at a time, so it tests unseen areas.

Baseline. A simple rule (the median price of the kecamatan or city) the model must beat.

Ensemble. The average of three models: XGBoost, extra trees and ridge regression.

Text score. A number from the listing's title and description, learned from the full descriptions of other listings.

SHAP effect. How much, on average, an input moves the estimate. It describes the model, not the housing market.

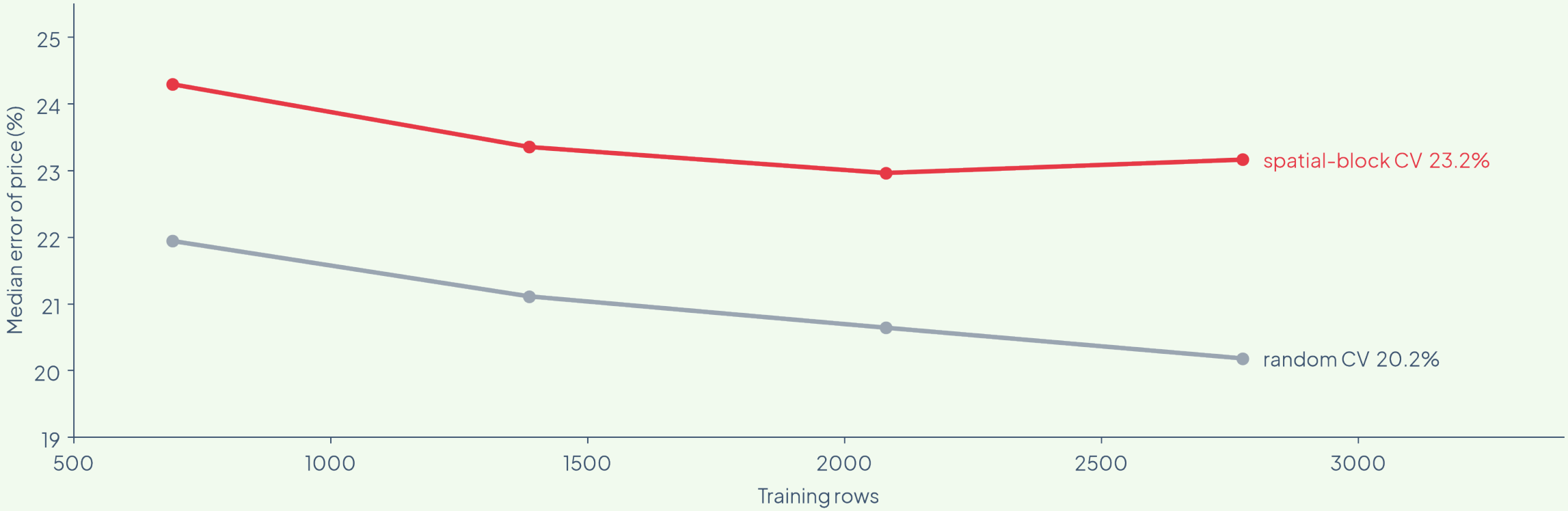
Kecamatan. An Indonesian sub-district; the unit of the price map.

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More rows help known areas, but under spatial testing the error has flattened since about 2,000 rows

Random CV falls from 21.9% to 20.2%; spatial CV from 24.3% to 23.2%, with most of the gain before 2,000 rows.

Median error of price against training rows, 3 repeats (%)



Source: learning_curve.csv. Thin number: 3 repeats.

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About

A model of Jabodetabek house asking prices, built from public listing pages, OpenStreetMap, BIG boundaries and BNPB flood hazard. Aggregates and estimates are published; listing-level data is not. Nothing here is a valuation or financial advice.

Author

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Related

Live map and estimator: https://harga-rumah-jabodetabek.netlify.app
Code (public showcase): github.com/rlukafiardi/jabodetabek-house-price-showcase

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